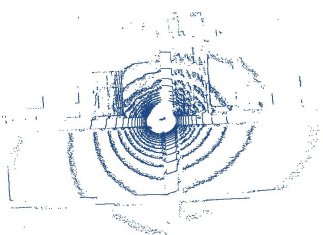


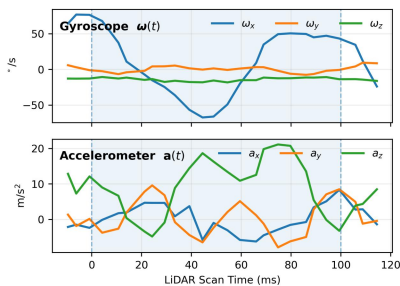
Sensor Input & Preprocessing



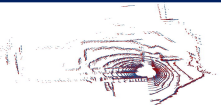
Raw Point Cloud



IMU Measurements

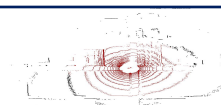


IMU-assisted LiDAR Deskewing Motion Compensation

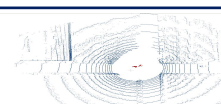


Preprocessing

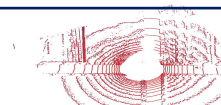
1. Range Filtering Distance Constraint



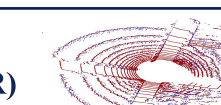
2. ROI & Self-body Filtering PassThrough & CropBox



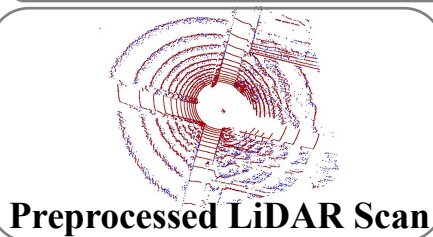
3. Voxel Downsampling VoxelGrid



4. Outlier Rejection Radius Outlier Removal (ROR)



LiDAR-Inertial Odometry(LIO)



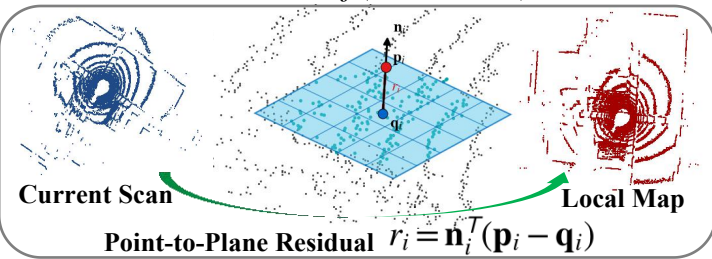
Preprocessed LiDAR Scan



IMU (High Rate)

IMU Propagation (State Prediction)

$$\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{imu})$$

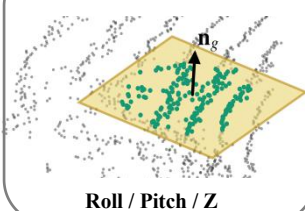


Current Scan

Local Map

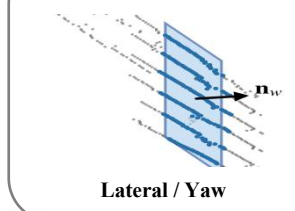
$$\text{Point-to-Plane Residual } r_i = \mathbf{n}_i^T (\mathbf{p}_i - \mathbf{q}_i)$$

Ground Constraint



Roll / Pitch / Z

Wall Constraint



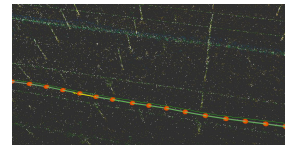
Lateral / Yaw

Iterated Error-State Kalman Filter(IESKF)

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}(\mathbf{z}_{lidar} - h(\hat{\mathbf{x}}_{k|k-1}))$$

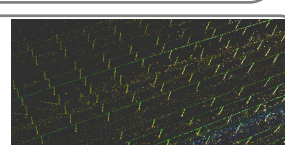
LIO Pose Output

$$\mathbf{T}_{WL} = (\mathbf{R}, \mathbf{p})$$



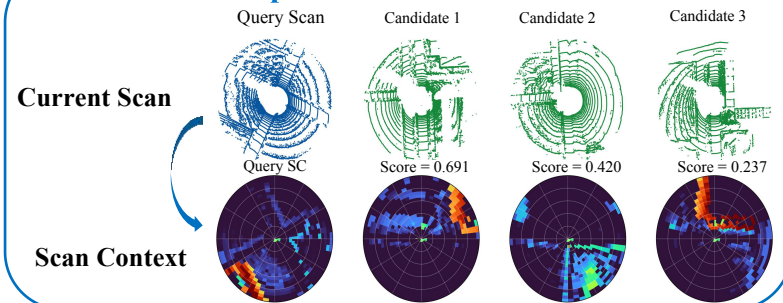
Local Map Update

(KeyFrame / Submap)



Loop Closure & Backend Optimization

Loop Candidate retrieval



Current Scan

Query Scan

Candidate 1

Candidate 2

Candidate 3

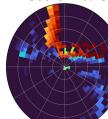
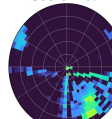
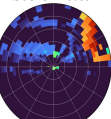
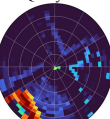


Score = 0.691

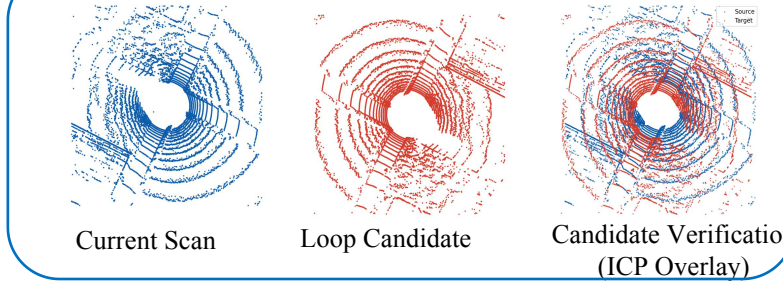
Score = 0.420

Score = 0.237

Scan Context



Candidate Verification

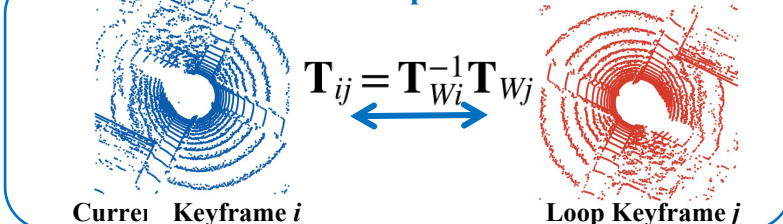


Current Scan

Loop Candidate

Candidate Verification
(ICP Overlay)

Verified Loop Constraint



Current Keyframe i

Loop Keyframe j

$$\mathbf{T}_{ij} = \mathbf{T}_{Wi}^{-1} \mathbf{T}_{Wj}$$

Pose Graph Optimization



Optimized Trajectory & Global Map

